

WSLC-499 ETHICAL HACKING BOOTCAMP (COHORT-17)

NAME-> Veer Pratap Singh Rathore

DATE->30/03/26

Project 3: Create a Directory Hierarchy

TASK 1: Create a Directory Hierarchy

```
CyberLab
├── Recon
│   ├── nmap
│   └── nikto
├── Exploitation
│   ├── metasploit
│   └── payloads
├── Reports
│   ├── screenshots
│   └── logs
├── Tools
│   ├── wordlists
│   └── scripts
└── CaseStudies
    ├── CaseStudy1.txt
    └── CaseStudy2.txt
```

1. Create main folder named CyberLab

```
(kali@kali)-[~]
$ mkdir CyberLab
```

2. Inside CyberLab, create all subfolders exactly as shown in the structure above.

```
(kali@kali)-[~/CyberLab]
$ mkdir -p Recon/{nmap,nikto} Exploitation/{metasploit,payloads} Reports/{screenshots,logs} Tools/{wordlists,scripts} CaseStudies
```

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3. Inside the CaseStudies folder, create two text files: CaseStudy1.txt , CaseStudy2.txt

```
(kali㉿kali)-[~/CyberLab/CaseStudies]  
$ touch CaseStudy1.txt CaseStudy2.txt
```

Screenshot of Directory Structure

```
(kali㉿kali)-[~]  
$ tree CyberLab  
CyberLab  
├── CaseStudies  
│   ├── CaseStudy1.txt  
│   └── CaseStudy2.txt  
├── Exploitation  
│   ├── metasploit  
│   └── payloads  
├── Recon  
│   ├── nikito  
│   └── nmap  
├── Reports  
│   ├── logs  
│   └── screenshots  
└── Tools  
    ├── scripts  
    └── wordlists  
  
14 directories, 2 files
```

Explanation:

- In this task, a structured directory hierarchy was created using mkdir command.
- Subdirectories were generated using brace expansion for efficiency.
- Two text files were created inside the CaseStudies folder to store real-world cyber attack case studies.

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TASK 2: Permission Hardening

Screenshot of permission command output

A screenshot of a terminal window with a dark background. The prompt is (kali㉿kali)-[~]. The first command is \$ chmod 700 CyberLab/CaseStudies. The second command is \$ ls -ld CyberLab/CaseStudies. The output of the second command is drwx----- 2 kali kali 4096 Apr 11 08:54 CyberLab/CaseStudies.

```
(kali㉿kali)-[~]  
$ chmod 700 CyberLab/CaseStudies  
  
(kali㉿kali)-[~]  
$ ls -ld CyberLab/CaseStudies  
drwx----- 2 kali kali 4096 Apr 11 08:54 CyberLab/CaseStudies
```

- ✓ Applied permission 700 on the CaseStudies folder
- ✓ Owner: read, write, execute
- ✓ Group: no access
- ✓ Others: no access
- ✓ This ensures that only the owner has full control over the CaseStudies directory, improving security.

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Case Studies:

Case Study 1: MGM Resorts Cyberattack

```
(kali@kali)-[~/CyberLab/CaseStudies]
$ cat CaseStudy1.txt
Name of the Incident: MGM Resorts Cyberattack
Year of Incident: 2023
Target Organization: MGM Resorts International
Type of Attack: Ransomware and Social Engineering

Why It Happened (Cause/Vulnerability):
The attackers didn't hack the system directly at first. Instead, they used social engineering. They called the company's IT help desk and pretended to be an employee. Because of weak identity verification, they were able to get access to internal systems.

What Happened After (Impact):
The attack caused major disruption. Many of MGM's systems stopped working, including hotel check-ins, slot machines, and digital room keys. Guests had a lot of problems during their stay. The company also lost millions of dollars, and some customer data may have been exposed.

How the Organization Responded:
MGM quickly shut down affected systems to stop the spread of the attack. They brought in cybersecurity experts to investigate and fix the issue. Over time, they restored their systems and informed customers about the situation.

Lessons Learned / Prevention Methods:
This incident shows how dangerous social engineering can be. Companies should train employees to verify identities properly. Using multi-factor authentication (MFA) and stricter security checks can help prevent similar attacks in the future.
```

- ◆ **Name of Incident:** MGM Resorts Cyberattack
- ◆ **Year of Incident:** 2023
- ◆ **Target Organization:** MGM Resorts International
- ◆ **Type of Attack:** Ransomware and Social Engineering

Why It Happened (Cause/Vulnerability)

- The attackers did not directly hack the systems initially. Instead, they used social engineering techniques.
- They contacted the company's IT help desk and impersonated authorized employees.
- Due to weak identity verification procedures, the attackers successfully gained access to internal systems.

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Impact of the Attack

- The cyberattack caused major operational disruptions across MGM Resorts.
- Hotel check-in services, slot machines, and digital room key systems stopped functioning properly.
- Customers faced inconvenience during their stay due to service interruptions.
- The organization suffered significant financial losses.
- Sensitive customer information may have been exposed during the incident.

Response by the Organization

- MGM Resorts immediately shut down affected systems to contain the attack.
- Cybersecurity experts were engaged to investigate the incident and restore operations.
- The organization gradually recovered critical services and resumed normal business activities.
- Customers were informed about the situation and advised regarding necessary precautions.

Lessons Learned / Prevention Methods

- ✓ Organizations should strengthen identity verification processes.
- ✓ Employees must receive regular training on recognizing social engineering attacks.
- ✓ Multi-factor authentication (MFA) should be implemented across critical systems.
- ✓ Continuous monitoring and rapid incident response mechanisms are essential.
- ✓ Strong access control policies can help reduce the impact of future attacks.

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Case Study 2: Change Healthcare Ransomware Attack

```
(kali@kali)-[~/CyberLab/CaseStudies]
$ cat CaseStudy2.txt
Name of the Incident: Change Healthcare Ransomware Attack
Year of Incident: 2024
Target Organization: Change Healthcare (UnitedHealth Group)
Type of Attack: Ransomware

Why It Happened (Cause/Vulnerability):
The attackers were able to get into the system using stolen login credentials. One major issue was that multi-factor authentication was not properly used on some important systems, which made it easier for attackers to gain access.

What Happened After (Impact):
This attack had a serious impact on healthcare services in the United States. Many pharmacies were unable to process prescriptions, and hospitals faced delays. It also raised concerns about patient data security. The company suffered financial losses and service interruptions.

How the Organization Responded:
The company shut down affected systems and worked with cybersecurity experts and law enforcement. They slowly restored their services and took steps to improve their security systems.

Lessons Learned / Prevention Methods:
This case shows the importance of strong login security. Organizations should always use multi-factor authentication, monitor systems regularly, and protect sensitive data carefully to avoid such attacks.
```

- ◆ **Name of Incident:** Change Healthcare Ransomware Attack
- ◆ **Year of Incident:** 2024
- ◆ **Target Organization:** Change Healthcare (UnitedHealth Group)
- ◆ **Type of Attack:** Ransomware

Why It Happened (Cause/Vulnerability)

- Attackers gained access using stolen login credentials.
- One of the major weaknesses was the absence of multi-factor authentication on certain important systems.
- Weak access security controls made it easier for attackers to infiltrate the network.

Impact of the Attack

- Healthcare services across the United States experienced serious disruptions.
- Many pharmacies were unable to process prescriptions on time.
- Hospitals and healthcare providers faced delays in claims processing.
- The incident raised concerns regarding the confidentiality and security of patient data.
- The company experienced financial losses and interruptions in business operations.

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Response by the Organization

- Change Healthcare shut down affected systems to limit further damage.
- The organization collaborated with cybersecurity professionals and law enforcement agencies.
- Recovery efforts were initiated to restore essential healthcare services.
- Security controls were reviewed and strengthened to prevent similar incidents in the future.

Lessons Learned / Prevention Methods

- ✓ Multi-factor authentication should be mandatory for all critical systems.
- ✓ Organizations should regularly monitor network activities for suspicious behavior.
- ✓ Strong password policies and credential management practices must be enforced.
- ✓ Sensitive healthcare data should be protected using robust security measures.
- ✓ A well-defined incident response plan is necessary to minimize the impact of cyberattacks.